

QUANTUM FIELD THEORY

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ABSTRACT. Quantum Field Theory (QFT) provides the essential theoretical framework that unifies quantum mechanics and special relativity. These notes explore the foundational concepts and mathematical machinery required to describe systems with infinite degrees of freedom, where particles are understood as localized excitations of underlying fields. Beginning with the transition from classical to quantum field formalisms, the text develops the canonical and path integral methods of quantization. Core topics include the analysis of free and interacting fields, the derivation of scattering amplitudes, and the implementation of diagrammatic perturbation theory. Symmetries, conservation laws, and the techniques necessary to handle regularized infinities are examined systematically. Ultimately, this material establishes the universal language used to analyze complex, multi-particle interactions across both high-energy and many-body physics.

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1. SIMPLE HARMONIC OSCILLATORS

1.1. Introduction

The idea of thinking particles are waves such that they obey the Schrödinger equation is known as the **first quantization**.

Definition 1.1. The **first quantization** is that particles behave like waves.

The reciprocal of this phenomenon is known as the **second quantization**.

Definition 1.2. The **second quantization** is that waves can behave like particles.

1.2. Mass on a Spring

Consider a mass on a spring.

Its Schrödinger equation is:

$$\left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + \frac{1}{2} K x^2 \right) \psi = E \psi, \quad (1)$$

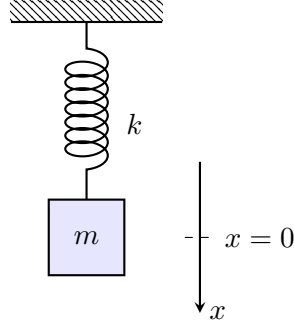
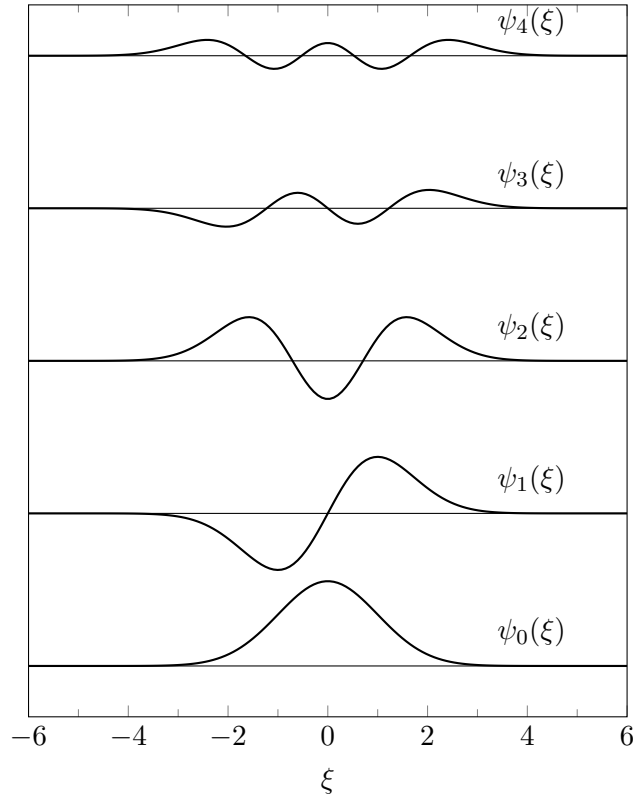


Figure 1: Mass-spring system at equilibrium.

and the respective solution is:

$$\psi_n(\xi) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi \hbar} \right)^{1/4} H_n(\xi) e^{-\xi^2/2}. \quad (2)$$

From eq.(2), $H_n(\xi)$ is a Hermite polynomial and $\xi = \sqrt{m\omega/\hbar}x$. When we plot the solutions, they appear to be wave-like.

Figure 2: **Fig. 2.2** Eigenfunctions of the simple harmonic oscillator.

However, there eigenvalues indicate discrete quantization, or *particle-like* behaviour.

$$E_n = \left(n + \frac{1}{2} \right) \hbar \omega, \quad (3)$$

where $\omega = \sqrt{K/m}$. When $n = 0$, $E_n = \frac{1}{2}\hbar\omega$, which we call **zero-point energy**. We can showcase the same result without the use of Hermite polynomials.

We start by expressing the Hamiltonian in terms for the SHO as:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2. \quad (4)$$

Factoring out the Hamiltonian, we examine the product of the two operator terms. Since $\hat{x}$ and $\hat{p}$ do not commute, we must maintain their relative order during expansion:

$$\frac{1}{2}m\omega^2 \left(\hat{x} - \frac{i}{m\omega}\hat{p} \right) \left(\hat{x} + \frac{i}{m\omega}\hat{p} \right) = \frac{1}{2}m\omega^2 \left(\hat{x}^2 + \frac{i}{m\omega}\hat{x}\hat{p} - \frac{i}{m\omega}\hat{p}\hat{x} - \frac{i^2}{m^2\omega^2}\hat{p}^2 \right) \quad (5)$$

$$= \frac{1}{2}m\omega^2 \left(\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} + \frac{i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x}) \right) \quad (6)$$

$$= \frac{1}{2}m\omega^2\hat{x}^2 + \frac{\hat{p}^2}{2m} + \frac{i\omega}{2}[\hat{x}, \hat{p}]. \quad (7)$$

Using the canonical commutation relation $[\hat{x}, \hat{p}] = i\hbar$, the final term evaluates to:

$$\frac{i\omega}{2}(i\hbar) = -\frac{\hbar\omega}{2}. \quad (8)$$

Substituting this back into Eq. (7) yields the relationship to the original Hamiltonian $\hat{H}$:

$$\frac{1}{2}m\omega^2 \left(\hat{x} - \frac{i}{m\omega}\hat{p} \right) \left(\hat{x} + \frac{i}{m\omega}\hat{p} \right) = \hat{H} - \frac{\hbar\omega}{2}, \quad (9)$$

which gives our desired factorized expression:

$$\hat{H} = \frac{1}{2}m\omega^2 \left(\hat{x} - \frac{i}{m\omega}\hat{p} \right) \left(\hat{x} + \frac{i}{m\omega}\hat{p} \right) + \frac{\hbar\omega}{2}. \quad (10)$$

However, $\hat{x}$ and $\hat{p}$ do not commute, so when we multiply the factored form, we obtain:

$$\frac{1}{2}m\omega^2\hat{x}^2 + \frac{\hat{p}^2}{2m} + \frac{i\omega}{2}[\hat{x}, \hat{p}], \quad (11)$$

and using the canonical commutation relation:

$$[\hat{x}, \hat{p}] \equiv \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar, \quad (12)$$

the expression simplifies to:

$$\frac{1}{2}m\omega^2\hat{x}^2 + \frac{\hat{p}^2}{2m} - \frac{\hbar\omega}{2} = \hat{H} - \frac{\hbar\omega}{2}. \quad (13)$$

Note that:

$$[\hat{x}, \hat{p}] \equiv \hat{x}\hat{p} - \hat{p}\hat{x} = i\hbar \quad (14)$$

From (10), we realise those terms are of use, so we can label them as $\hat{a}^\dagger$ and $\hat{a}$:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega}\hat{p} \right), \quad (15)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega}\hat{p} \right). \quad (16)$$

The definitions of $\hat{a}$ and $\hat{a}^\dagger$ can be inverted to give expressions for $\hat{x}$ and $\hat{p}$. We begin with the forward definitions:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right), \quad (17)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right). \quad (18)$$

To isolate $\hat{x}$, we add Eq. (17) and Eq. (18), which cancels out the momentum $\hat{p}$ term:

$$\hat{a} + \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left[\left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) + \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \right] \quad (19)$$

$$= \sqrt{\frac{m\omega}{2\hbar}} (2\hat{x}) \quad (20)$$

$$= \sqrt{\frac{4m\omega}{2\hbar}} \hat{x} = \sqrt{\frac{2m\omega}{\hbar}} \hat{x}. \quad (21)$$

Inverting the prefactor yields the final expression for position:

$$\hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger). \quad (22)$$

To isolate $\hat{p}$, we subtract Eq. (18) from Eq. (17), which cancels out the position $\hat{x}$ term:

$$\hat{a} - \hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left[\left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) - \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \right] \quad (23)$$

$$= \sqrt{\frac{m\omega}{2\hbar}} \left(\frac{2i}{m\omega} \hat{p} \right) \quad (24)$$

$$= \frac{2i}{m\omega} \sqrt{\frac{m\omega}{2\hbar}} \hat{p} = 2i \sqrt{\frac{m\omega}{2\hbar m^2 \omega^2}} \hat{p} = i \sqrt{\frac{4}{2\hbar m \omega}} \hat{p} = i \sqrt{\frac{2}{\hbar m \omega}} \hat{p}. \quad (25)$$

Multiplying both sides by $-i$ (since $-i \cdot i = 1$) and inverting the prefactor yields the final expression for momentum:

$$\hat{p} = -i \sqrt{\frac{\hbar m \omega}{2}} (\hat{a} - \hat{a}^\dagger). \quad (26)$$

Substituting eq. (26) into eq. (4), we obtain:

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right). \quad (27)$$

In this refined Hamiltonian, the active ingredient is $\hat{a}^\dagger \hat{a}$. If $\hat{a}^\dagger \hat{a}$ has an eigenstate $|n\rangle$ with eigenvalue n , then $\hat{H}$ will also have an eigenstate $|n\rangle$ with an eigenvalue $\hbar\omega(n + 1/2)$, such that we've recovered the eigenvalues of the simple harmonic oscillator.

This is simply achieved by doing the following:

$$n = \langle n | \hat{a}^\dagger \hat{a} | n \rangle = |\hat{a} | n \rangle|^2 \geq 0. \quad (28)$$

The next thing to show is that n takes only integer values.

Definition 1.3. The **number operator** is defined as $\hat{n} = \hat{a}^\dagger \hat{a}$.

By this definition, we have:

$$\hat{n}|n\rangle = n|n\rangle. \quad (29)$$

We can also express the Hamiltonian again by the following equations:

$$\hat{H} = \hbar\omega \left(\hat{n} + \frac{1}{2} \right), \quad (30)$$

$$\hat{H}|n\rangle = \left(n + \frac{1}{2} \right) \hbar\omega|n\rangle, \quad (31)$$

Example 1.4. This example investigates the properties of the excited state created by applying the creation operator to an energy eigenstate, written as $\hat{a}^\dagger|n\rangle$. We can analyze this state by applying the number operator $\hat{n}$:

$$\hat{n}\hat{a}^\dagger|n\rangle = \hat{a}^\dagger\hat{n}\hat{a}^\dagger|n\rangle. \quad (32)$$

By substituting the standard commutation relation $[\hat{a}, \hat{a}^\dagger] = 1$ (rewritten as $\hat{a}\hat{a}^\dagger = 1 + \hat{a}^\dagger\hat{a}$), we can simplify the expression:

$$\hat{n}\hat{a}^\dagger|n\rangle = (n+1)\hat{a}^\dagger|n\rangle. \quad (33)$$

The example above defines $\hat{a}^\dagger$ as the **raising operator**.

Example 1.5. This example investigates the properties of the state created by applying the annihilation operator to an energy eigenstate, written as $\hat{a}|n\rangle$. We can analyze this state by applying the number operator $\hat{n}$:

$$\hat{n}\hat{a}|n\rangle = \hat{a}^\dagger\hat{a}\hat{n}|n\rangle. \quad (34)$$

Using the commutation relation from equation 2.11, we substitute $\hat{a}^\dagger\hat{a} = \hat{a}\hat{a}^\dagger - 1$, which yields:

$$\hat{n}\hat{a}|n\rangle = (n-1)\hat{a}|n\rangle. \quad (35)$$

The example above defines $\hat{a}$ as the **lowering operator**. We can further call $\hat{a}^\dagger$ the **creation operator** and $\hat{a}$ the **annihilation operator**.

1.3. A Trivial Generalization

Consider a set of N uncoupled simple harmonic oscillators. Let

$$\hat{H} = \sum_{k=1}^N \hat{H}_k, \quad (36)$$

express the sum of the individual Hamiltonians. We express the individual Hamiltonian as:

$$\hat{H}_k = \frac{\hat{p}_k^2}{2m_k} + \frac{1}{2}m_k\omega_k^2\hat{x}_k^2. \quad (37)$$

We again, define the creation and annihilation operators, which obey the following commutation rules:

$$\begin{cases} [\hat{a}_k, \hat{a}_q] &= 0, \\ [\hat{a}_k^\dagger, \hat{a}_q^\dagger] &= 0, \\ [\hat{a}_k, \hat{a}_q^\dagger] &= \delta_{kq}. \end{cases} \quad (38)$$

Therefore, $\bar{H}$ becomes:

$$\hat{H} = \sum_{k=1}^N \hbar\omega_k \left(\hat{a}_k^\dagger\hat{a}_k + \frac{1}{2} \right). \quad (39)$$

The general state of the system $|n_1, n_2, \dots, n_N\rangle$ is known as the **occupation number representation**. This can be written as:

$$|n_1, n_2, \dots, n_N\rangle = \frac{1}{\sqrt{n_1! n_2! \dots n_N!}} (\hat{a}_1^\dagger)^{n_1} (\hat{a}_2^\dagger)^{n_2} \dots (\hat{a}_N^\dagger)^{n_N} |0, 0, \dots, 0\rangle, \quad (40)$$

and more formally,

$$|\{n_k\}\rangle = \prod_k \frac{1}{\sqrt{n_k!}} (\hat{a}_k^\dagger)^{n_k} |0\rangle. \quad (41)$$

1.4. Phonons

We begin with a 1D chain of N identical atoms, each of mass m , arranged in a ring (periodic boundary conditions) with an equilibrium spacing a . The displacement of the j -th atom from its equilibrium position is denoted by $\hat{x}_j$, and its momentum is $\hat{p}_j$.

Because adjacent atoms are connected by chemical bonds modeled as springs with spring constant K , the potential energy depends on the relative displacement between neighbors, $(\hat{x}_{j+1} - \hat{x}_j)$. The total Hamiltonian operator representing the energy of the system is:

$$\hat{H} = \sum_j \left[\frac{\hat{p}_j^2}{2m} + \frac{1}{2} K (\hat{x}_{j+1} - \hat{x}_j)^2 \right] \quad (42)$$

This Hamiltonian is difficult to solve because the potential energy mixes terms from different sites (j and $j+1$). To solve it, we must decouple the degrees of freedom by transforming from real space to momentum space using a Discrete Fourier Transform (DFT).

We re-express the position and momentum operators as a linear combination of collective plane waves characterized by a wavevector k :

$$\hat{x}_j = \frac{1}{\sqrt{N}} \sum_k \hat{x}_k e^{ikja} \quad (43)$$

$$\hat{p}_j = \frac{1}{\sqrt{N}} \sum_k \hat{p}_k e^{ikja} \quad (44)$$

The inverse transforms (expressing the momentum-space operators in terms of real-space operators) are:

$$\hat{x}_k = \frac{1}{\sqrt{N}} \sum_j \hat{x}_j e^{-ikja} \quad (45)$$

$$\hat{p}_k = \frac{1}{\sqrt{N}} \sum_j \hat{p}_j e^{-ikja} \quad (46)$$

Because the atoms form a closed loop, the j -th atom is identical to the $(j+N)$ -th atom. This imposes periodic boundary conditions:

$$e^{ikja} = e^{ik(j+N)a} \implies e^{ikNa} = 1 \quad (47)$$

This forces the wavevector k to take discrete, quantized values:

$$k = \frac{2\pi m}{Na} \quad (48)$$

where m is an integer restricted to the first Brillouin zone: $-\frac{N}{2} < m \leq \frac{N}{2}$.

When summing phase factors over all lattice sites, the phases interfere destructively and sum to zero unless the exponent is zero. This yields the identity:

$$\sum_j e^{i(k+k')ja} = N\delta_{k,-k'} \quad (49)$$

where $\delta_{k,-k'}$ is the Kronecker delta (equals 1 if $k' = -k$, and 0 otherwise).

In real space, the canonical commutation relation is $[\hat{x}_j, \hat{p}_{j'}] = i\hbar\delta_{jj'}$. We must verify that our new Fourier variables preserve quantum mechanics:

$$[\hat{x}_k, \hat{p}_{k'}] = \frac{1}{N} \sum_j \sum_{j'} e^{-ikja} e^{-ik'j'a} [\hat{x}_j, \hat{p}_{j'}] \quad (50)$$

Substitute $[\hat{x}_j, \hat{p}_{j'}] = i\hbar\delta_{jj'}$ into the double sum to collapse the j' index:

$$[\hat{x}_k, \hat{p}_{k'}] = \frac{i\hbar}{N} \sum_j e^{-i(k+k')ja} \quad (51)$$

Applying our orthogonality identity reduces this expression to:

$$[\hat{x}_k, \hat{p}_{k'}] = i\hbar\delta_{k,-k'} \quad (52)$$

We substitute the Fourier expansions into the sum of $\hat{p}_j^2$:

$$\sum_j \hat{p}_j^2 = \sum_j \left(\frac{1}{\sqrt{N}} \sum_k \hat{p}_k e^{ikja} \right) \left(\frac{1}{\sqrt{N}} \sum_{k'} \hat{p}_{k'} e^{ik'ja} \right) \quad (53)$$

Rearranging the summation signs and factoring out the constants gives:

$$\sum_j \hat{p}_j^2 = \frac{1}{N} \sum_k \sum_{k'} \hat{p}_k \hat{p}_{k'} \sum_j e^{i(k+k')ja} \quad (54)$$

Using the identity $\sum_j e^{i(k+k')ja} = N\delta_{k,-k'}$, the N terms cancel out, and the Kronecker delta forces $k' = -k$, collapsing the k' sum:

$$\sum_j \hat{p}_j^2 = \sum_k \hat{p}_k \hat{p}_{-k} \quad (55)$$

First, we look at the difference term $(\hat{x}_{j+1} - \hat{x}_j)$ using the Fourier expansion:

$$\hat{x}_{j+1} - \hat{x}_j = \frac{1}{\sqrt{N}} \sum_k \hat{x}_k (e^{ik(j+1)a} - e^{ikja}) = \frac{1}{\sqrt{N}} \sum_k \hat{x}_k e^{ikja} (e^{ika} - 1) \quad (56)$$

Squaring this difference and summing over j creates a triple sum:

$$\sum_j (\hat{x}_{j+1} - \hat{x}_j)^2 = \frac{1}{N} \sum_k \sum_{k'} \hat{x}_k \hat{x}_{k'} (e^{ika} - 1) (e^{ik'a} - 1) \sum_j e^{i(k+k')ja} \quad (57)$$

Once again, the sum over j becomes $N\delta_{k,-k'}$, which collapses the k' sum by setting $k' = -k$:

$$\sum_j (\hat{x}_{j+1} - \hat{x}_j)^2 = \sum_k \hat{x}_k \hat{x}_{-k} (e^{ika} - 1) (e^{-ika} - 1) \quad (58)$$

Now, we expand the product of the two bracketed expressions:

$$(e^{ika} - 1)(e^{-ika} - 1) = e^0 - e^{ika} - e^{-ika} + 1 = 2 - (e^{ika} + e^{-ika}) \quad (59)$$

Using Euler's identity $\cos(\theta) = \frac{e^{i\theta} + e^{-i\theta}}{2}$, this simplifies to:

$$2 - 2\cos(ka) = 2(1 - \cos(ka)) \quad (60)$$

Using the trigonometric half-angle identity $1 - \cos(\theta) = 2\sin^2\left(\frac{\theta}{2}\right)$, we arrive at:

$$2 \cdot 2\sin^2\left(\frac{ka}{2}\right) = 4\sin^2\left(\frac{ka}{2}\right) \quad (61)$$

Substituting this back into the potential energy expression yields:

$$\sum_j (\hat{x}_{j+1} - \hat{x}_j)^2 = \sum_k \hat{x}_k \hat{x}_{-k} \left(4\sin^2\left(\frac{ka}{2}\right) \right) \quad (62)$$

Combining both simplified terms back into the original Hamiltonian gives:

$$\hat{H} = \sum_k \left[\frac{1}{2m} \hat{p}_k \hat{p}_{-k} + \frac{1}{2} m \omega_k^2 \hat{x}_k \hat{x}_{-k} \right] \quad (63)$$

where we have defined the dispersion relation ω_k^2 as:

$$\omega_k^2 = \frac{4K}{m} \sin^2\left(\frac{ka}{2}\right) \quad (64)$$

The atoms are now completely uncoupled. Each wavevector k behaves as an independent harmonic oscillator mode.

To solve this system using standard quantum mechanics, we define creation ($\hat{a}_k^\dagger$) and annihilation ($\hat{a}_k$) operators for each mode k :

$$\hat{a}_k = \sqrt{\frac{m\omega_k}{2\hbar}} \left(\hat{x}_k + \frac{i}{m\omega_k} \hat{p}_k \right) \quad (65)$$

$$\hat{a}_k^\dagger = \sqrt{\frac{m\omega_k}{2\hbar}} \left(\hat{x}_{-k} - \frac{i}{m\omega_k} \hat{p}_{-k} \right) \quad (66)$$

Note: Because $\hat{x}_j$ and $\hat{p}_j$ are Hermitian physical observables, their Fourier transforms satisfy $\hat{x}_k^\dagger = \hat{x}_{-k}$ and $\hat{p}_k^\dagger = \hat{p}_{-k}$. This is why the indices flip sign in the creation operator.

These operators satisfy the standard harmonic oscillator commutation relations:

$$[\hat{a}_k, \hat{a}_{k'}^\dagger] = \delta_{kk'}, \quad [\hat{a}_k, \hat{a}_{k'}] = 0, \quad [\hat{a}_k^\dagger, \hat{a}_{k'}^\dagger] = 0 \quad (67)$$

Inverting the equations allows us to express our momentum-space operators in terms of ladder operators:

$$\hat{x}_k = \sqrt{\frac{\hbar}{2m\omega_k}} (\hat{a}_k + \hat{a}_{-k}^\dagger) \quad (68)$$

$$\hat{p}_k = -i\sqrt{\frac{m\hbar\omega_k}{2}} (\hat{a}_k - \hat{a}_{-k}^\dagger) \quad (69)$$

Substituting the equations for $\hat{x}_k$ and $\hat{p}_k$ back into the decoupled Hamiltonian leads to the intermediate form:

$$\hat{H} = \sum_k \frac{\hbar\omega_k}{2} \left(\hat{a}_k \hat{a}_k^\dagger + \hat{a}_{-k}^\dagger \hat{a}_{-k} \right) \quad (70)$$

Since we sum over all positive and negative values of k , renaming the index variable from $-k$ to k for the second term is mathematically valid:

$$\hat{H} = \sum_k \frac{\hbar\omega_k}{2} \left(\hat{a}_k \hat{a}_k^\dagger + \hat{a}_k^\dagger \hat{a}_k \right) \quad (71)$$

Finally, we use the commutator relation $\hat{a}_k \hat{a}_k^\dagger = \hat{a}_k^\dagger \hat{a}_k + 1$ to rewrite the equation in its standard, final form:

$$\hat{H} = \sum_{k=1}^N \hbar\omega_k \left(\hat{a}_k^\dagger \hat{a}_k + \frac{1}{2} \right) \quad (72)$$

REFERENCES

- [1] T. Lancaster and S. J. Blundell, *Quantum Field Theory for the Gifted Amateur*, Oxford University Press, 2014.